

Maths Teacher may call and ask you the answers of these or same type of questions to verify your knowledge.

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Weekly Test

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Keep it up.

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Class :- 10th

Set A

Sub :- Maths (211)

Roll no :-

Ans 1:- C) 560 ✓

Ans 2:- B) $18a^3b^2$ ✓

Ans 3:- A) Fig A = $(a+b)^2 = a^2 + 2ab + b^2$, Fig B = $(a-b)^2 = a^2 - 2ab + b^2$,
Fig C = $(a+b)(a-b) = a^2 - b^2$ ✓

Ans 4:- B) (i) & (iv) ✓

Ans 5:- D) All are in non-standard form ✓

Ans 6:- C) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ✓

Ans 7:- B) 2, 6, 10 ✓

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Ans 8:- C) $S_n = \frac{n}{2} (a+1)$ ✓

Ans 9:- A) $4a^2 + 12ab + 9b^2$ ✓

Ans 10:- C) 3 ✓

Ans 11:-

Ans 11:-

i) False ✓

ii) True ✓

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iii) False ✓

iv) True ✓

Ans 12:-

i) Using $(a+b)^2 = a^2 + 2ab + b^2$:

$$\left(\frac{x}{3}\right)^2 + 2\left(\frac{x}{3}\right)(1) + (1)^2 = \frac{x^2}{9} + \frac{2x}{3} + 1$$

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ii)

$$\left(x + \frac{4}{3}\right)\left(x + \frac{3}{4}\right)$$

$\Rightarrow$ Using $(x+a)(x+b) = x^2 + (a+b)x + ab$:

$$x^2 + \left(\frac{4}{3} + \frac{3}{4}\right)x + \left(\frac{4}{3} \times \frac{3}{4}\right)$$

$$= x^2 + \left(\frac{16+9}{12}\right)x + 1 = x^2 + \frac{25}{12}x + 1 \quad \checkmark$$

Q.13

i) $7x + 2 = 9$

$\Rightarrow 7x = 9 - 2$

$\Rightarrow 7x = 6$

$\Rightarrow x = \frac{6}{7} \quad \checkmark$

ii) $\frac{3y}{2} - 3 = 9$

$\Rightarrow \frac{3y}{2} = 9 + 3$

$\Rightarrow \frac{3y}{2} = 12$

$\Rightarrow 3y = 24$

$\Rightarrow y = 8 \quad \checkmark$

Q.14

i) $3x^2 + 7x + 2 = 0$

$\Rightarrow a = 3, b = 7, c = 2$

$D = b^2 - 4ac = (7)^2 - 4(3)(2) = 49 - 24 = 25 \quad \checkmark$

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Since $D > 0$, the roots are real and distinct (unequal) ✓

ii) $25x^2 + 20x + 4 = 0$

⇒ Here, $a = 25, b = 20, c = 4$

$D = b^2 - 4ac = (20)^2 - 4(25)(4) = 400 - 400 = 0$ ✓

Since $D = 0$, the roots are equal and real

Q.15

Ans i) • First term $a = 2$, common

difference $d = 3$, last term $l = 59$

• Find number of terms n :

$l = a + (n-1)d \Rightarrow 59 = 2 + (n-1)3$

• Sum

$S_n = \frac{n}{2} (a+l) = \frac{20}{2} (2+59) = 10 \times 61 = \underline{610}$ ✓

Ans ii) • First term $a = 7$, last term $l = 994$

• Number of terms $n = \frac{994 - 7}{7} + 1 = 142$

• Sum

$S_{142} = \frac{142}{2} (7+994) = 71 \times 1001 =$

71071 ✓

4

Ans iii) n^{th} term of $5, 9, 13, 17 \dots$ is

⇒ • $a = 5, d = 4$

• $a_n = a + (n-1)d = 5 + (n-1)4 = 5$

• $4n + 1$ (or $1 + 4n$) ✓

Ans iv) For a quadratic equation

$ax^2 + bx + c = 0, a \neq 0$ will not have any real roots when the Discriminant $D = b^2 - 4ac$ is

less than zero negative < 0 ✓

Q.19
⇒

Q.16

i) Real and equal roots occur when $D=0$. In the table, $D=0$ for equation P.

⇒ A) P ✓

ii) No real roots occur when $D < 0$. In the table, $D = -4$ for equation R.

⇒ C) R ✓

iii) Two distinct real roots occur when the Discriminant is positive ($D > 0$)

• Equation Q has $D = 1 > 0$

• Equation S has $D = 25 > 0$

⇒ B) Q and S ✓

iv) Looking at the table for equation S ($2x^2 + 7x + 3 = 0$), the listed value under Value of Discriminant (D) is 25

⇒ C) 25 ✓

v) If the discriminant of a quadratic equation is negative, then its roots are:

⇒ C) Not real ✓

Q.17

⇒ Using the standard identity for the cube of a binomial:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 \quad \checkmark \quad 1/2$$

multiply and find not just write the formula

Q.18

⇒ $x, x+2$ and $x+4$

$$(x) + (x+2) + (x+4) = 36$$

$$3x + 6 = 36$$

$$3x = 36 - 6$$

$$3x = 30 \Rightarrow x = 10$$

First integer = $x = 10$

10, 12 and 14 ✓

Second integer = $x+2 = 12$

Third integer = $x+4 = 14$

Q.19

⇒ Let the unit's digit be y and the ten's digit be x .

According to the problem:

$$x = 2y + 2$$

$$x \times y = 24 \Rightarrow (2y + 2)y = 24 \Rightarrow 2y^2 + 2y - 24 = 0$$

Divide by 2

$$y^2 + y - 12 = 0 \Rightarrow (y + 4)(y - 3) = 0$$

Since digits must be positive, $y = 3$.

Find x :

$$x = 2(3) + 2 = 8$$

The no. is $10x + y = 10(8) + 3 = 83$ ✓

Q.20. Each term of an AP whose first term is a and common difference is d , is doubled. Is the resulting pattern an AP? If so find its first term and common difference.

⇒ Original AP: $a, a + d, a + 2d, a + 3d, \dots$

Doubled Terms: $2a, 2(a + d), 2(a + 2d), 2(a + 3d), \dots$

⇒ $2a, 2a + 2d, 2a + 4d, 2a + 6d, \dots$

Difference between consecutive terms: $(2a + 2d) - 2a = 2d$.

Since the difference is constant ($2d$),

Yes, the resulting pattern is an AP.

• 1st Term: $2a$

• Common difference: $2d$ ✓

Q.21

⇒ Let Robert's present age be x years.

Then Asha's present age is $(x + 5)$ years.

Five years ago:

• Robert's age = $x - 5$

• Asha's age = $(x + 5) - 5 = x$

$$x = 2(x - 5) \Rightarrow x = 2x - 10 \Rightarrow x = 10$$

Robert's present age - 10 years ✓

2

Asha's present age - 15 years ✓

Q.22

⇒ Let breadth $b = x$ m. Then length

$$l = (x+7) \text{ m.}$$

$$\text{Area} = l \times b = 144 \Rightarrow x(x+7) = 144$$

$$(x+16)(x-9) = 0 \Rightarrow x = 9 \text{ (since } x > 0)$$

$$\text{Breadth} = 9 \text{ m, Length} = 16 \text{ m}$$

✓

✓

Q.23 F

⇒ First 3 digit no. leaving 1 as the remainder when divided by 4: 101 ($100+1$)

Last 3 digit no.: 997 ($996+1$)

AP series: 101, 105, 109, ..., 997

$$(a = 101, d = 4, l = 997)$$

Find no. of terms (n):

$$997 = 101 + (n-1)4 \Rightarrow 896 = 4(n-1) \Rightarrow n-1 = 224 \Rightarrow n = 225$$

Sum (S_n):

$$S_{225} = \frac{225}{2} (101 + 997) = \frac{225}{2} (1098) = 225 \times 549 = 123,525$$

$$\Rightarrow 123,525 \text{ ✓}$$

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Q.24

⇒

Let Aj's actual marks be: x

Given condition

$$9(x+10) = x^2 \Rightarrow 9x+90 = x^2 \Rightarrow x^2 - 9x - 90 = 0$$

$$(x-15)(x+6) = 0 \Rightarrow x = 15 \text{ since marks } \geq 0$$

$$\Rightarrow 15 \text{ ✓}$$

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